

Ansible cho mô hình HA Proxy và Web Server

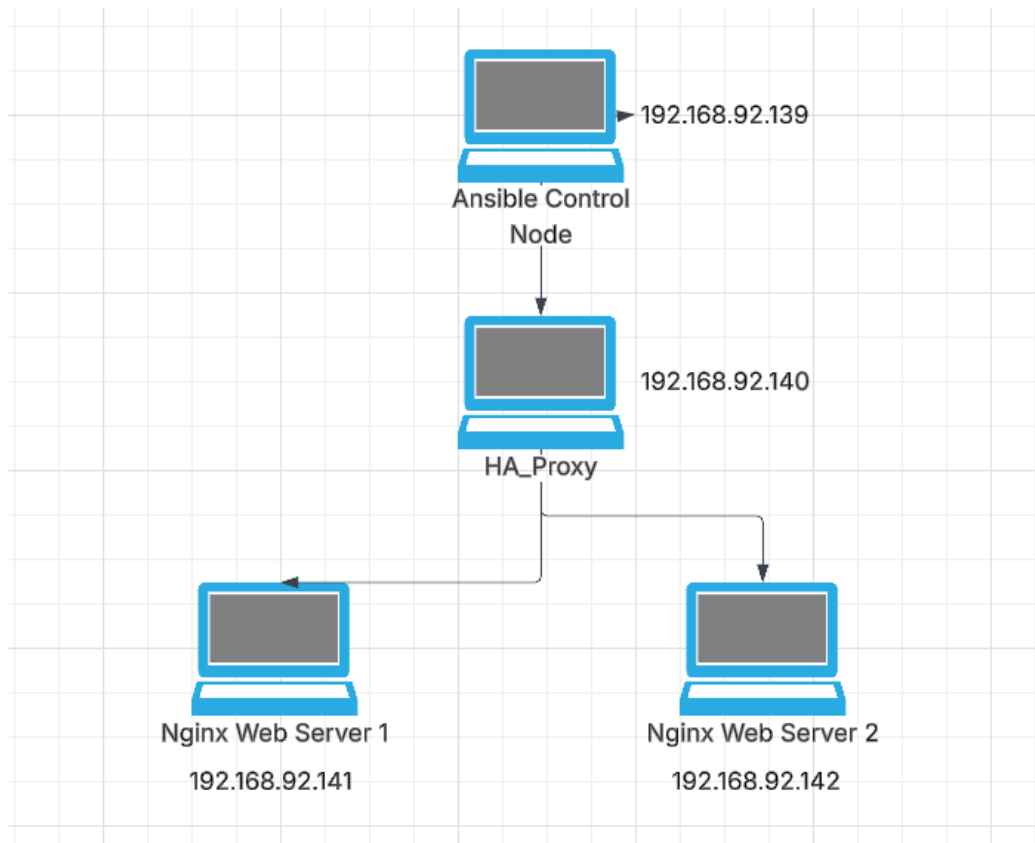
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Ngày 16 tháng 6 năm 2026

Mục lục

I	Mô hình	2
II	Kiểm tra kết quả	3
II.1	Chạy Play book thành công	3
II.2	Cân bằng tải của Proxy	3
II.3	Verify HAProxy Config	5
II.4	Simulate Failure	6
III	Intermediate+	6
III.1	Add Variable	6
III.2	Add Stats Page	7
III.3	Add Rolling Deployment	8

I Mô hình



Hình 1: Mô hình Ansible HA Proxy

II Kiểm tra kết quả

II.1 Chạy Play book thành công

```
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ ansible-playbook -i inventory.ini site.yml

PLAY [Configure Web Servers] *****

TASK [Gathering Facts] *****
ok: [web1]
ok: [web2]

TASK [nginx : Install nginx] *****
ok: [web1]
ok: [web2]

TASK [nginx : Deploy index page] *****
ok: [web2]
changed: [web1]

TASK [nginx : Ensure nginx running] *****
ok: [web1]
ok: [web2]

RUNNING HANDLER [nginx : restart nginx] *****
changed: [web1]

PLAY [Configure Load Balancer] *****

TASK [Gathering Facts] *****
ok: [lb01]

TASK [haproxy : Install haproxy] *****
ok: [lb01]

TASK [haproxy : Deploy HAProxy config] *****
ok: [lb01]

TASK [haproxy : Ensure haproxy running] *****
ok: [lb01]

PLAY RECAP *****
lb01      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web1      : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Hình 2: Chạy Playbook thành công

II.2 Cân bằng tải của Proxy

Liên tục truy cập `http://192.168.92.140` thì thấy được cân bằng tải lúc thì hiện Web1, lúc hiện Web2

```
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ curl http://192.168.92.140
<!DOCTYPE html>
<html>
<head>
  <title>Ansible HAProxy Lab</title>
</head>
<body>

<h1>Hello from web2</h1>

<p>Managed by Ansible</p>

<p>Server IP: 192.168.92.142</p>

</body>
</html>
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ curl http://192.168.92.140
<!DOCTYPE html>
<html>
<head>
  <title>Ansible HAProxy Lab</title>
</head>
<body>

<h1>Hello from web1</h1>

<p>Managed by Ansible</p>

<p>Server IP: 192.168.92.141</p>

</body>
</html>
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ curl http://192.168.92.140
<!DOCTYPE html>
<html>
<head>
  <title>Ansible HAProxy Lab</title>
</head>
<body>

<h1>Hello from web2</h1>

<p>Managed by Ansible</p>

<p>Server IP: 192.168.92.142</p>

</body>
</html>
```

Hình 3: Cân Bằng tải HA Proxy

II.3 Verify HAProxy Config

```
vanquyen@VanQuyen:~$ ssh vanquyen@192.168.92.140
vanquyen@192.168.92.140's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun 16 02:30:06 AM UTC 2026

System load:  0.0          Processes:           218
Usage of /:   55.7% of 9.75GB Users logged in:           1
Memory usage: 9%          IPv4 address for ens33: 192.168.92.140
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Jun 16 02:03:37 2026 from 192.168.92.139
vanquyen@HAProxy:~$ sudo haproxy -c -f /etc/haproxy/haproxy.cfg
Configuration file is valid
vanquyen@HAProxy:~$
```

Hình 4: Verify HA Proxy

```
vanquyen@HAProxy:~$ systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-06-16 01:55:42 UTC; 36min ago
     Docs: man:haproxy(8)
           file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 2322 (haproxy)
    Status: "ready"
      Tasks: 3 (limit: 4543)
     Memory: 39.4M (peak: 39.9M)
        CPU: 1.932s
   CGroup: /system.slice/haproxy.service
           └─2322 /usr/sbin/haproxy -ss -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
             └─2324 /usr/sbin/haproxy -ss -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock

Jun 16 01:55:42 HAProxy systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
Jun 16 01:55:42 HAProxy haproxy[2322]: [NOTICE] (2322) : New worker (2324) forked
Jun 16 01:55:42 HAProxy haproxy[2322]: [NOTICE] (2322) : Loading success.
Jun 16 01:55:42 HAProxy systemd[1]: Started haproxy.service - HAProxy Load Balancer.
Jun 16 01:55:42 HAProxy haproxy[2324]: [WARNING] (2324) : Server web_back/web1 is DOWN, reason: Layer4 connection problem, info: "Connection refused", check duration: 0ms, 1 active and 0 backup servers left, 0
Jun 16 02:00:14 HAProxy haproxy[2324]: [WARNING] (2324) : Server web_back/web2 is DOWN, reason: Layer4 connection problem, info: "General socket error (Network is unreachable)", check duration: 0ms, 0 active and 0 backup servers left, 0 sessions requested, 0 total in queue
Jun 16 02:00:14 HAProxy haproxy[2324]: [ALERT] (2324) : backend 'web_back' has no server available!
Jun 16 02:00:18 HAProxy haproxy[2324]: [WARNING] (2324) : Server web_back/web2 is UP, reason: Layer4 check passed, check duration: 0ms, 1 active and 0 backup servers online, 0 sessions requested, 0 total in queue
Jun 16 02:03:07 HAProxy haproxy[2324]: [WARNING] (2324) : Server web_back/web1 is UP, reason: Layer4 check passed, check duration: 0ms, 2 active and 0 backup servers online, 0 sessions requested, 0 total in queue
vanquyen@HAProxy:~$
```

Hình 5: Trạng thái HA Proxy

II.4 Simulate Failure

Tiến hành tắt nginx trên máy Web 2 và truy cập dịch vụ web trên HA Proxy. HA proxy sẽ trở về Web1

```
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ curl http://192.168.92.140
<!DOCTYPE html>
<html>
<head>
  <title>Ansible HAProxy Lab</title>
</head>
<body>

<h1>Hello from web1</h1>

<p>Managed by Ansible</p>

<p>Server IP: 192.168.92.141</p>

</body>
</html>
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$
```

Hình 6: Simulate Failure

III Intermediate+

III.1 Add Variable

Mục tiêu: Sau này muốn thay đổi Port thì chỉ cần thay đổi 1 chỗ chứ không cần phải đi thay từng nơi

```
cd ~/ansible-haproxy-lab
mkdir -p group_vars
# Tao File loadbalancers.yml
vim group_vars/loadbalancers.yml(lưu ý tên File phải
    giống Group trong inventory.ini)

haproxy_frontend_port: 80
haproxy_backend_port: 80

# Sua file template HAProxy
bind *:80 -> bind *:{ haproxy_frontend_port }
{% for host in groups['webservers'] %}
    server {{ host }} {{ hostvars[host]['ansible_host']
    }}:80 check
```

```
{% endfor %} -> {% for host in groups['webservers'] %}
    server {{ host }} {{ hostvars[host]['ansible_host']
        }}:{{haproxy_backend_port}} check
{% endfor %}
```

Kiểm tra :

```
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ ansible -i inventory.ini loadbalancers -m debug -a "var=haproxy_frontend_port"
lb01 | SUCCESS => {
  "haproxy_frontend_port": 80
}
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ ansible -i inventory.ini loadbalancers -m debug -a "var=haproxy_backend_port"
lb01 | SUCCESS => {
  "haproxy_backend_port": 80
}
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ ansible-playbook -i inventory.ini site.yml --syntax-check
playbook: site.yml
```

Hình 7: Kết quả Add Variables

III.2 Add Stats Page

```
global
    log /dev/log local0
    log /dev/log local1 notice
    daemon

defaults
    mode http
    timeout connect 5s
    timeout client 50s
    timeout server 50s

frontend http_front
    bind *:{{ haproxy_frontend_port }}
    default_backend web_back

backend web_back
    balance roundrobin

{% for host in groups['webservers'] %}
    server {{ host }} {{ hostvars[host]['ansible_host'] }}:{{ haproxy_backend_port }} check
{% endfor %}

listen stats
    bind *:8404
    stats enable
    stats uri /stats
    stats refresh 10s
```

Hình 8: Add State

```
vanquyen@AnsibleControlNode:~/ansible-haproxy-lab$ ansible -i inventory.ini loadbalancers -m shell -a "ss -tulnp | grep 8404" -b -K
BECOME password:
lb01 | CHANGED | rc=0 >>>
tcp LISTEN 0 4096 0.0.0.0:8404 0.0.0.0:* users:((("haproxy",ptd=4504,fd=10))
```

Hình 9: Kiểm tra Add State thành công không

III.3 Add Rolling Deployment

```
- - -
- name: Configure Web Servers
  hosts: webservers
  serial: 1
  become: yes

  roles:
    - nginx

- name: Configure Load Balancer
  hosts: loadbalancers
  become: yes

  roles:
    - haproxy
```

Hình 10: Add Rolling Deployment

Giúp cài đặt lần lượt từng Web Server : Web Server 1 rồi đến WebServer 2

```
root@ansiblecontrolnode: /root/.ansible/ansible-playbook -i inventory.ini site.yml
PLAY [Configure Web Servers] *****
TASK [Gathering Facts] *****
ok: [web1]

TASK [nginx : Install nginx] *****
ok: [web1]

TASK [nginx : Deploy index page] *****
ok: [web1]

TASK [nginx : Ensure nginx running] *****
ok: [web1]

PLAY [Configure Web Servers] *****
TASK [Gathering Facts] *****
ok: [web2]

TASK [nginx : Install nginx] *****
ok: [web2]

TASK [nginx : Deploy index page] *****
ok: [web2]

TASK [nginx : Ensure nginx running] *****
ok: [web2]

PLAY [Configure Load Balancer] *****
TASK [Gathering Facts] *****
ok: [lb01]

TASK [haproxy : Install haproxy] *****
ok: [lb01]

TASK [haproxy : Deploy HAProxy config] *****
ok: [lb01]

TASK [haproxy : Ensure haproxy running] *****
ok: [lb01]

PLAY RECAP *****
web1 : ok=4 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
web2 : ok=4 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```

Hình 11: Cài đặt lần lượt từng WebServer